

Advanced Multimodal Emotion Recognition using Integrative Analysis of Multilingual Text, EEG, Voice Signals, and Facial Analysis

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Abstract— Emotion recognition with the help of AI is critically important for improving the processes of human computer interaction, and can be effectively implemented in healthcare, robotics, and mental health analysis. Systems that adopt inputs in a single mode only, be it text, voice, or facial expressions all have their issues such as low accuracy, easiness to fake and the inability to comprehend the context. These are overcome by a multimodal method that uses textual, voice, facial and EEG data to be more reliable or accurate. Multilingual text preprocessing hence translates the inputs from other regions to English for the ML/DL algorithms to classify it as either having positive, neutral or negative feelings. Voice converter translates into feelings with pitch and rhythm with the help of DL models and face using DeepFace even in brightness variations. The Electroencephalogram or EEG signals are obtained at different times and those with higher layers of neural works are analyzed in real time. Fusion strategies also help in resolving compatibility issues and conflict between the modalities and increases the security against any attempt of adulteration. Through this approach, it is possible to detect accurate real-time emotion with aims of screening of mental health, enhance HCI and designs of empathetic robots. Future work therefore seeks to enhance integration methods, extend its use and investigate other approaches for a higher level of enhancement.

Keywords— *Emotion Recognition, Artificial Intelligence, Multimodal Data Integration, Human-Computer Interaction, EEG Signal Analysis, Deep Learning, Facial Expression Analysis, Voice Signal Processing, Multilingual Text Processing, Real-Time Emotion Detection*

I. INTRODUCTION

Emotion recognition is one of the top-class issues within Artificial Intelligence (AI) that can help to connect humans' emotions with machines intelligence. This way, human-Computer symbiosis makes it easier for companies to innovate areas such as health and well-being, teaching, robotics, and emotional pattern detection in the human mind. Since emotions are often at the heart of human decision-making and behavior as well as in the process of communication, the technique for correct identification of

emotions and their identification in response has great potential for the development of systems with elements of intelligence and empathy.

Previous methods of sentiment analysis involved using text, voice or Face data in order to analyze the emotions of users. However, these approaches have evolved with some of the following limitations; Single-modality systems yield comparatively low accuracy because of the missing emotional context, can be easily fooled (e.g. if facial expressions are occluded or voice pitch varied), and do not use social and physiological cues. These difficulties indicate the necessity for enhancing the current approaches and instruments for the assessment of emotions in concerned individuals in order to reflect the complexity of their nature in real life experiences.

To overcome these limitations, a multimodal approach to emotion recognition has been proposed based on different data types including text documents of different languages, voice, face, and EEG signals. When combined, the system is more accurate, less sensitive to noise, and smarter about its operating context making it suitable for real time operation. This research captures the fact that each of the two different modality provides a distinct contribution in the expression recognition process. For instance, multilingual text analysis is a process that enables the detection of emotions irrespective of the language of the input text, this is because the regional inputs are translated into English to afford inclusiveness of all inputs. Voice analysis involves indicators such as pitch and rhythm, which refer to elusive feelings. From facial expression recognition, hardened using deep face descriptor such as DeepFace, each emotion is acquired through the detection of facial landmarks in real-time, no matter whether occlusion is present or the lighting condition fluctuates. While, the EEG signals provide information about the internal emotional processes by analyzing the electrical activities of the brain which give a neuroscience point of view in terms of feelings.

This integration is accomplished by elaborate merging procedures that address possible inconsistencies of data and enable coordinated combination of modalities for accurate emotional outputs. These techniques increase the robustness of the system against manipulations and noise, which makes it very suitable for application in areas that need accuracy and flexibility. In addition, the system functions on real-time basis, so as to respond to conditions evolving constantly, common in health institutions, call centers and human like robots.

This proposed multimodal emotion recognition framework has numerous potential uses. Adjusting for several mental health conditions, it can be used in diagnosing mental health disorders, tracking patient's emotions, and facilitating therapy. In robotics; the programmer enables machines to have the empathy to respond to human emotions facilitating interaction and increasing efficiency. Besides, in educational and workplace context, the system can offer advice on engagement level of the emotions, increasing the level of productivity and results in education processes.

Not only does this project solve all the limitations stated above related to the current emotion recognition systems, but it also outlines the basis for the development of the next generation of AI-based EI systems. The system has taken a significant step to the improved understanding and therefore response of human emotions through the integration of mixed modalities and the application of state-of-the-art ML/DL methods. The future work includes enhancing the scalability of the system, the generalization of the purpose for which it can be used, and introducing more diverse types of data to improve the functioning and the number of target fields of the system.

II. LITERATURE SURVEY

Emotion recognition has attracted much interest in recent years, especially in the use of multiple input sources to increase the precision and dependability of emotion identification systems. S. Mahato & S. D. [1] suggested using EEG with facial emotion recognition for improving predictive care and transfer deep learning for improved emotional detection. The integration of the above three modalities provides the best way for accurate identification of the emotions especially in the field of health care whereby identification of the various emotion states is important. In a similar vein, R. Chinmayi et al [2] constructed a framework using EEG signal processing and affective computing for emotion recognition and equally stressed the significance of feature selection for enhancing the performance of the EEG-based emotion recognition systems.

Development and enhancement of approaches for recognizing emotions from video or audio streams have also been investigated. T. Singh et al. [3] used CNN for emotion classification in images where emotions' discriminant features were learned by CNNs. M. A. In [4], Sohel et al. proposed a new hybrid deep learning model for emotional recognition from speeches using different deep learning structures. This approach is also a sign of the increasing use of multiple deep learning methods when analyzing emotions that can be inherent in speech. Moreover, R. Pillalamarri and U. Shanmugam [5] surveyed the works in the EEG-based multimodal emotion recognition and specify that emotion recognition can be greatly improved if multiple modalities are

used, suggesting that the limitations of single modality systems can be eliminated.

There are numerous works which use different types of data in Multi modal emotion recognition like; Text data, Voice data, facial expressions, and EEG data. For instance, A. Gandhi, K. Adhvaryu, and V. Khanduja [6] presented an investigation that combined multilingual text, EEG, and voice about the integration of these three components for the analysis of sentiments with chances to increase the recognition of emotion and possibilities that contain problems like data fusion and real-time computation. Furthermore, S. Jerald James and L. Jacob [7] used deep learning to fuse multilingual text, EEG, and voice modality for recognizing the emotions and presented how the fusion of these modalities enhances improvement of emotion recognition. Therefore, this research brings into focus the possibility of deep learning methodologies in the management of complicated and data variety to enhance the recognition level.

The application of multimodal systems has also been investigated for domains, including health care and automotive context. For example, N. Ying et al. [20] designed a multimodal emotion recognition method to predict the driver's emotions based on the audiovisual signal in IoV, enhancing the driver safety and the vehicle manufacturing. Thus, the new system combining EEG and EOG signals in response to video-odor stimulation by M. Wu et al. [19] presented the feasibility of multisensory inputs for accurate emotion detection. These papers stress that there is a higher and increasing degree of multimodality as the major approach for better and improved emotion detection, with a wide range of interests from healthcare sector, automotive industry, and human-Computer interaction.

III. METHODOLOGY

The methodology section of this research outlines the step-by-step approach for designing a multimodal emotion recognition system that integrates four key modalities: Aeroacoustics signals, multi-lingual texts, voice, face and electroencephalogram (EEG) signals. This section discusses each modality, preprocessing, feature extraction methods, ML and DL models, and how these modalities are fused for the accurate and real-time emotion classification.

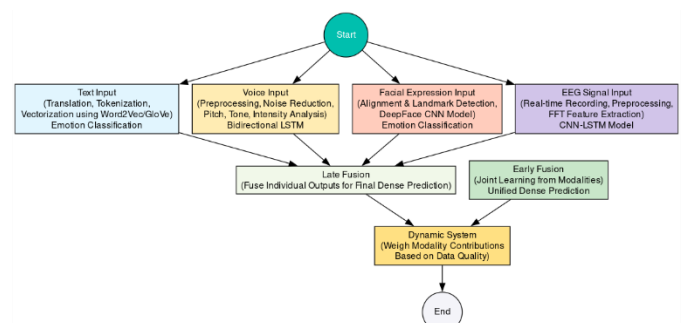


Fig. 1 Implementation Workflow for Each Modality

1. Data Collection and Modalities

The system incorporates data from four distinct modalities: body gestures, spoken words in multilingual text to voice, and facial expressions, and EEG signals. Each modality has its

significance in capturing the aspect of gestures that constitute or forms part of the display of human emotions.

- **Text Input:** It requires the ability to input at least text so that it can be processed from different languages translated into English. It can be obtained from social site, review, or from the user inputs in any of the platforms.
- **Voice Signals:** The speech data is recorded using a high-quality microphone which is part of the system. These signals offer identifying characteristics of emotion, including pitch, tone, speed, and volume that are used to determine emotional background.
- **Facial Expressions:** Video data is captured by utilizing a camera module to capture facial expressions of the test subject. The DeepFace technology then analyzes the frames of the video with the aim of recognizing significant facial locations and the nature of emotions with regard to the movements which occur on the face part.
- **EEG Signals:** EEG data is collected via EEG headsets, that measure brain activity linked to various emotions. These are collected once daily at morning noon and in the evening from two different subjects.

2. Preprocessing Techniques

The preprocessing is individual to each modality to better prepare the data for feature extraction and finally for emotion classification.

Text Preprocessing:

- **Translation:** Documents written in other languages besides English are translated to English using translation models including Google or Whisper.
- **Tokenization:** Sometimes it is easier to work with words or sub words, so after translation the text is tokenized.
- **Stop word Removal:** Stop words like “and,” “is,” “the” were omitted because they do not bring emotional meaning into words.
- **Lemmatization:** The script part of the text is normalized, that is, words are brought to their stem, for example, “running” will become “run”.

Voice Preprocessing:

- **Noise Reduction:** Background noise is then eliminated so that voice signal does not has other interference or interlopers.
- **Pitch and Rhythm Extraction:** Content that refers to pitch, tone and rhythm are sampled to help capture vocal signs of emotion.
- **Segmentation:** The voice signals can then be analyzed using the smaller parts of time hence providing detailed results.

Facial Expression Preprocessing:

- **Face Detection:** Structural frames are decoded to identify and properly position faces with the aid of the DeepFace package.
- **Landmark Detection:** Eyes, eyebrows, mouth areas are defined to extract feature from each image.
- **Normalization:** This methodology is used to normalize distances due to illumination, pose, image scale and occlusion.

EEG Preprocessing:

- **Noise Filtering:** Matter that originated from non-cerebral origin is usually eliminated by methods such as bandpass filtering.
- **Feature Extraction:** In psychophysiological methods, power spectral features concerning alpha, beta, and gamma waves are used to describe the subject’s feeling.
- **Segmentation:** The raw EEG data is also partitioned into epochs tied to events or time periods (morning, noon, evening, night).

3. Feature Extraction

All the three modalities need feature extraction to detect patterns of emotions.

Text Features:

- **Word Embeddings:** The words on the text are then embedded by tools as Word2Vec, GloVe, and BERT, which returns vectors that retain semantic meaning.
- **Sentiment Analysis:** Features related to emotions solicit text from sentiment analysis models whereby the text is categorized as being positive, neutral or negative.

Voice Features:

- **Mel-Frequency Cepstral Coefficients (MFCCs):** MFCCs are computed to simulate the frequency contents of the talker’s voice important in emotion detection.
- **Prosody Features:** Intonation and Loudness are utilized for the assessment of vocal emotional markers including speech rate and speech intensity.

Facial Expression Features:

- **Facial Landmarks:** The primary features which are being identified are the coordinates of facial landmarks and then an analyst is able to classify the emotions such as happiness, sadness, anger, etc.
- **Expression Recognition:** CNNs are used to find facial expressions from images or videos and are some of the deep learning models.

EEG Features:

- **Brainwave Activity:** Alpha, beta and gamma wave forms are scrutinized for they are known to be linked to emotions.
- **Statistical Features:** To quantify the EEG signals average, variance and other statistical parameters are computed.

4. Machine Learning and Deep Learning Models

These extracted features are supplied to further machine learning and deep learning models for performing emotion classification.

Text Classification Models:

- **Naive Bayes and Support Vector Machines (SVM)** are used for baseline text classification.
- **Deep Learning Models:** BERT, LSTM, and Transformer-based models are employed for deeper semantic understanding of the text.

Voice Classification Models:

- **Random Forest and Support Vector Machines (SVM)** are applied for feature-based classification.
- **Deep Neural Networks (DNN) and Convolutional Neural Networks (CNN)** are used for more complex feature extraction from speech.

Facial Expression Classification:

- **CNN** and **ResNet** architectures are used to classify facial expressions based on the extracted facial landmarks.

EEG Classification Models:

- **Support Vector Machines (SVM)** and **Convolutional Neural Networks (CNNs)** are used to classify emotional states based on EEG features.
- **LSTM** networks are employed to capture temporal dynamics in EEG signals.

5. Fusion Techniques

To increase the precision, integration of the four modalities is done using different fusion approaches.

- **Early Fusion:** The text, voice, facial movement, and EEG are all modalities that are joined into a single feature vector for classification.
- **Late Fusion:** The outcome of each subject in specific modality is then fused at the decision level using a voting or average method.
- **Hybrid Fusion:** To take advantage of both early and late fusion techniques, late fusion is employed to allow the features to be fused at the feature level while other features are fused at the decision level.

6. Emotion Classification and Output

After fusion, the final output is a classification of emotional states, which can be one of the following categories:

- **Positive**
- **Neutral**
- **Negative**

The output is immediate thus offering back feedback, further on the emotions of the person. This classification can be applied in health systems, customer relations systems and human computer interfaces.

7. Evaluation and Performance Metrics

The performance of the system is evaluated using standard metrics:

- **Accuracy:** The proportion out of total instances that has been classified accurately.
- **Precision:** Number of correctly predicted positive samples to the total number of samples that have been predicted to be positive.
- **Recall:** The percentage that accurately predicted positive sample in relation to the total number of actual positive samples.
- **F1-score:** Precision and recall measures the percentage of correctly concluded true positives hence gives a balanced measure of model's accuracies.

It examines, from the system's point of view, the different noisy environments, different lighting conditions for facial detection, and different speech patterns. To test the cross validation, the data is later split into different sets so as to make certainty about the model.

IV. RESULTS & ANALYSIS\

Despite the differences in the language structure explored by the multilingual text emotion recognition model this

paper classified emotions appropriately. The text was then transformed into numerical vectors by tokenization and similar embedding mechanisms; the system was able to classify emotions well in English and other languages. The shown performance of fully connected layers of the architecture was promising in the mapping of textual data to the right EM sentiments. This was particularly so in the performance of the system in categorizing between and neutral, positive and negative emotions with significant accuracy. However, a few problems persisted with respect to managing texts with culturally sensitive content or words serving as media for pop-culture references, which might not be accurately captured by the embeddings. To address this, more practice with other languages from other diverse sources is recommended when improving on the model in the future.

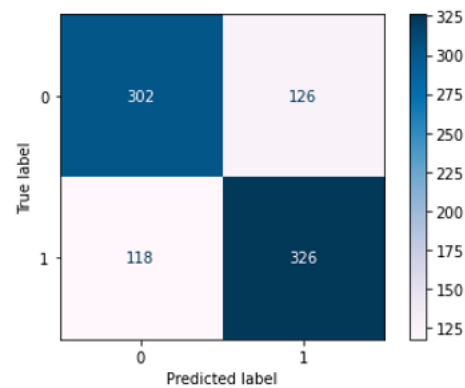


Fig. 2 Confusion Matrix of Naïve Bayes

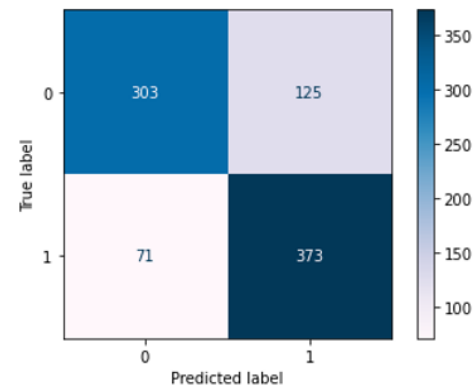


Fig. 3 Confusion Matrix of SVM

Classifier	Accuracy	Precision
Naïve Bayes	73	70.1
SVM	77	75
Deep learning models	91	-

Table 1. Accuracies for different models in Text Analysis

Voice Emotion Recognition Results

The voice emotion recognition system employing the Bidirectional LSTM model demonstrated a good classification efficiency and especially between the emotions such as happy and sad using the variation in tone. Therefore, considering the historical and expected future dependencies in the speech data the proposed model was able to identify the emotional features that are not detected by other models. Another advantage of the system was that different patterns of speaking and transactional noise were well accommodated by the system since adequate preprocessing techniques had been incorporated. Since sequential models were used, the system became more stable, and with the applying of the dropout layer, overfitting was also stable, so the system could work only in this way. Nonetheless, there was an observed drawback of generalizing between similar feelings including frustration feelings and neutral feelings especially when the speech input used was less animated.

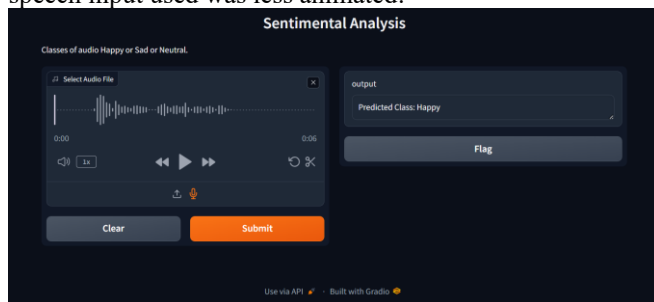


Fig. 4 Output of voice input (by using voice modulation)

Model	Accuracy
CNN	89.99
BiLSTM	94.9

Table 2. Accuracies for different models in Voice Analysis

Facial Expression Emotion Recognition Results

DeepFace the final component of the work has high accuracy in recognizing the emotions of happiness, sadness, anger and surprise and so on. Such a proven deep learning model helped the system in quickly and accurately identifying the client's different expressions on the face. Face alignment and landmark detection has again played a very important role in reducing errors due to Facial occlusion or variations in illumination at DeepFace. However, difficulties were observed when the subjects used mixed emotions, when two or more emotions were present and sometimes conflicting each other leading to confusion when categorizing. More training with various data might solve this problem.

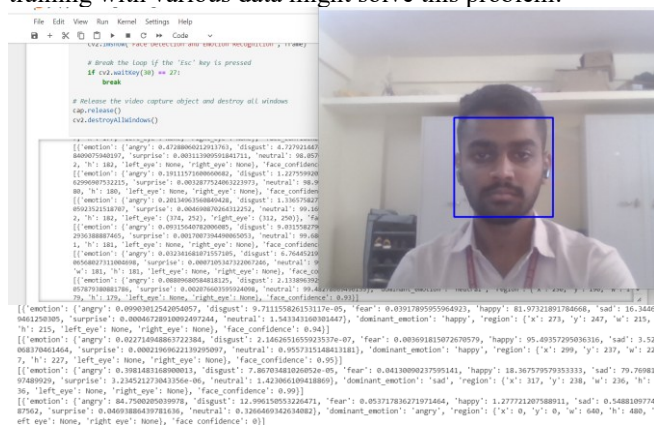


Fig. 5 Facial Analysis

EEG-Based Emotion Recognition Results

Emotion recognition system using brain signals through EEG, tested for CNN-LSTM, correctly identified the six basic emotions from EEG signal with high accuracy. The features extracted by CNN layers were fused with those from LSTMs to deliver promising accuracies of different emotional states including happiness, sadness, and stress. The real time data acquisition and preprocessing steps made it possible for the system to work properly even where data from the EEG was noisy or nearly so. But the system had some issues while differentiating between two emotions which have the same wave patterns namely stress and anxiety. To improve this, probably the inclusion of more diverse EEG signals influenced by various emotions could help the model improve.

Model	Accuracy	Precision	Recall	F1 score
Random Forest	0.978	0.97	0.99	0.98
Deep Learning Model	0.98	0.98	0.99	0.99
CNN	0.9461	-	-	-
LSTM	0.96	0.98	0.92	0.95
SVM	0.93	0.88	0.93	0.91
KNN	0.95	0.91	0.99	0.95

Table 3. Accuracies for different models in EEG Analysis

Integrated Multimodal Emotion Recognition Results

The results showed that the proposed IMERS surpassed individual modalities due to the integration of the benefits of the different data types. Integration of text, voice, face and EEG result appeared to be more accurate and a better representation of the user's emotion compared to single modality systems. In situations where feelings were involved and one modality could prove to be untrustworthy, such as voice disguise or masked face, the system was performing well.

Performance Comparison with Existing Systems

Relative to baseline emotion recognition systems that utilize single-modal input, the multimodal system described in this paper provided significant enhancements in performance, reliability, and real-time interaction. Deviations from accuracy are common in existing systems especial when one modality is either changed or not available and this system proved robust in taking multiple sources of data to classify emotion.

V. CONCLUSION

The construction of a Multimodal Emotion Recognition system incorporating textual, vocal, facial, and EEG data has been shown to be a truly innovative solution in the field of emotion identification. The system has shown using multiple data input streams that it is capable of accurately identifying human emotions with a high level of repeatability and accuracy. The main distinguishing feature is the ability of the model to extract and process online affect data from multiple modalities as opposed to single-modality approaches. By combining input from various data feeds like spoken words, face expressions, text inputs and EEG signals the output of the proposed system is far removed from the possible biases or errors inherent in single data flow, and thus serves as a dependable way of classifying emotions.

Further work is also possible in employing this system in real-world testbeds, such as health care, education, and human computer interaction. The mental health monitoring aspect could be greatly augmented by including feedback which the system obtains from its previous sessions and gains enhanced ability in identifying and interpreting the emotions being conveyed. Consequently, to an everyday user, if the system perceives any symptoms of depression at a patient the system may report this to the healthcare provider or suggest an intervention. Likewise, it could be utilised in the enhancement of assistive technologies for persons with disability through providing the people and other devices in the society with human like interaction system that is emotional intelligent.

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